

Analytical Summary

Linguistic Interpretability of Transformer-based Language Models: A Systematic Review

Based on López-Otal et al. (2025) • Prepared for an M.Sc. AI course deliverable

1. Main Problem and Why It Matters

Transformer-based pre-trained language models (PLMs) achieve strong performance on many NLP tasks, yet their internal computations remain difficult to interpret. The reviewed paper addresses a focused interpretability question: to what extent do Transformer PLMs encode linguistic phenomena—such as syntax, morphology, lexico-semantics, and discourse—in their internal representations, and where (layers, attention heads, neurons, vector subspaces) can this information be found?

This question matters for both science and engineering. Scientifically, it tests whether the inductive biases of Transformers support representations that resemble human linguistic generalizations. Practically, linguistic interpretability can help diagnose failure modes (e.g., errors with long-distance dependencies or negation), support multilingual transfer, and guide targeted interventions such as pruning, editing, or knowledge injection. The review also highlights a tension in the literature: some studies interpret linguistic-looking patterns as genuine generalizations, while others argue models may rely on shallow statistical cues.

2. What Are the Inputs and Outputs of the Model/System?

Because the paper is a systematic review, it does not propose a single new model. Instead, it organizes how prior studies treat a PLM plus an interpretability method as a ‘system’ that takes linguistic stimuli and produces evidence of encoded linguistic knowledge.

Typical Inputs	Typical Outputs
• A base, pre-trained Transformer LM (e.g., BERT, RoBERTa, mBERT) • Text stimuli (sentences, controlled minimal pairs, corpora) • Optional linguistic annotations (PoS, dependencies, semantic roles, discourse labels) • A chosen interpretability method (probing, attention analysis, attribution, ablation, etc.)	• Recoverability scores for linguistic features (e.g., probing accuracy) • Localization hypotheses (which layers/heads/neurons encode a phenomenon) • Visualizations (attention heatmaps, embedding geometry) • Error patterns, limitations, and practical takeaways

The review focuses on studies that inspect internal representations (layerwise embeddings, attention heads, neurons) of base PLMs without fine-tuning, and excludes purely black-box behavioral evaluations.

3. Data Used in the Review (Type, Source, Size)

The ‘data’ of the review is a curated corpus of research papers about linguistic interpretability in Transformer PLMs. Using a PRISMA-style process, the authors retrieved candidates via Google Scholar (2018–Oct 2024) and filtered them through multi-annotator screening, yielding 160 papers for detailed analysis.

Time window	2018 to October 2024 (reported search/selection period)
Primary sources	Google Scholar; ArXiv list cross-checked with ACL Anthology
Size of final corpus	160 research papers analyzed
Coverage highlights	Languages: English-centric (145/160 papers include English). Models: mainly BERT/RoBERTa variants and mBERT; fewer studies on modern decoder-only LLMs.

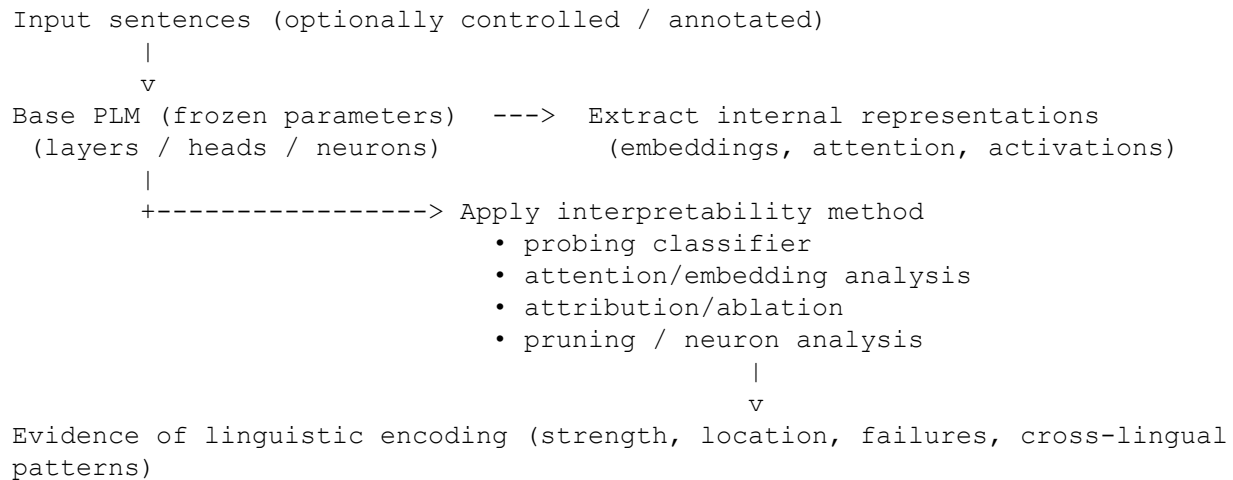
Across the surveyed papers, datasets vary widely: annotated corpora (PoS, dependencies, semantic roles, discourse labels), controlled stimuli (agreement, word order, negation), and multilingual benchmarks for cross-lingual sharing. The review synthesizes findings by linguistic discipline rather than endorsing a single benchmark.

4. Method (Explained Simply) + Schematic / Pseudocode

The review proposes a practical taxonomy of how researchers ‘extract’ linguistic information from PLMs: (1) feature attribution, (2) example-based explanations, (3) direct analysis of architectural elements (attention heads, neurons, embeddings, pruning), and (4) probing with external classifiers. Probing is the most frequent approach across the surveyed papers.

To standardize layer discussions, many studies use relative depth terms. For a 12-layer BERT-like encoder, ‘lower’ layers are roughly 1–3, ‘middle’ 4–7, and ‘upper’ 8–12. Architectural differences also matter: encoders attend to all tokens bidirectionally, while decoder-only models generate left-to-right; tokenization schemes differ as well (e.g., WordPiece vs BPE), which affects morphological analysis.

Schematic: typical linguistic interpretability workflow



Pseudocode: minimal layerwise probing experiment

```
for layer in PLM.layers:
    E = [PLM(s).hidden_states[layer] for s in dataset]
    probe = LogisticRegression() # or small MLP
    probe.fit(E_train, y_train)
    score[layer] = probe.evaluate(E_test, y_test)

report(best_layer = argmax(score), score_curve = score)
```

A central caveat is correlation vs causation: extracting a linguistic feature does not prove the PLM uses it during inference. The review notes a trend toward causal probing and intervention-based tests to address this gap.

5. Main Results, Limitations, and Future Directions

5.1 Key findings by linguistic level

Syntax (most studied):

Syntactic information is often recoverable from PLMs (dependencies, agreement, grammatical relations). Yet layerwise localization is disputed: many works highlight middle layers in BERT-like models, while others find important information in lower layers or distributed across the network. Reported weaknesses include low-frequency agreement forms, sensitivity to word order in some settings, and long-distance relations in certain languages.

Lexico-semantics:

Semantic properties (polysemy, similarity/relatedness, argument structure, scalar properties) are widely reported, but sometimes less robust than syntax. Difficult phenomena (negation, compositionality) show mixed evidence. Lower layers are often described as more lexical, while higher layers may reflect more contextual semantics depending on model and pre-training objective.

Morphology:

Morphological and morphosyntactic features (number, gender, tense, agreement) can be extracted, but tokenization complicates analysis. Some studies place morphosyntax in middle layers (similar to syntax) and report partial cross-lingual sharing in multilingual PLMs, while others emphasize language-dependent variation.

Discourse (least studied):

A small set of papers suggests discourse information is partially encoded (often in intermediate layers), but coverage is limited and models may struggle with the semantics of discourse relations.

5.2 Quantitative trends and methodological observations

In the authors' tally, syntax is the dominant linguistic focus (113 papers), followed by (lexico-)semantics (82), morphology (38), and discourse (6). Methodologically, probing is most common (117 papers), followed by analysis of architectural elements (58), with feature attribution and example-based approaches appearing less frequently (16 each).

- English dominates; many low-resource languages remain under-studied.
- BERT/RobERTa (and mBERT) are common targets because internal-access methods scale poorly to very large or closed models.
- Probing is often combined with perturbations or attention/embedding analyses to strengthen conclusions.
- Recoverability does not imply causal use; causal probes and interventions are increasingly important.

5.3 Limitations and ideas for continuation

Key limitations include (i) skewed coverage toward English and older, accessible encoder models; (ii) methodological ambiguity in probing (task learning, instability, and metric limitations); and (iii) interpretability cost at scale for contemporary LLMs, which pushes research toward black-box prompting and reduces transparency.

Future directions: extend internal-access studies to newer open LLMs; use stronger causal evaluation (interventions, counterfactual editing); expand typologically diverse multilingual coverage; and connect interpretability outcomes to downstream robustness and targeted model improvement.

Reference

López-Otal, M., Gracia, J., Bernad, J., Bobed, C., Pitarch-Ballesteros, L., Anglés-Herrero, E. (2025). Linguistic Interpretability of Transformer-based Language Models: a systematic review. ACM Computing Surveys (manuscript).